DATE : 19.10.2026

DT/NT : DT

LESSON : AWS

SUBJECT : LAUNCH TEMPLATE

BATCH : B 424-020DE

AWS-DEVOPS

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EC2 Launch Template

What Does a Launch Template Store?

AMI

Amazon Linux 2023 (OS image)

Instance Type

t3.micro (CPU / memory)

Storage

EBS root volume (for example gp3)

Network and Security

Security Group, key pair, optional network settings

Automation and Metadata

User data, IAM role, instance and volume tags

One launch template → Consistent EC2 launches

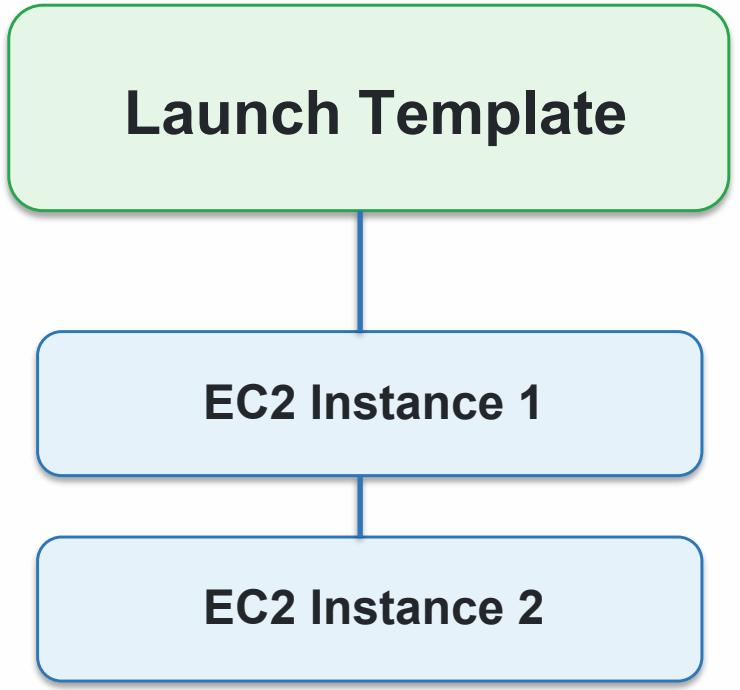
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EC2 Launch Template

What Is an EC2 Launch Template?

- A reusable set of EC2 launch parameters.
- Launch instances consistently and repeatably.
- Create versions as your application changes.

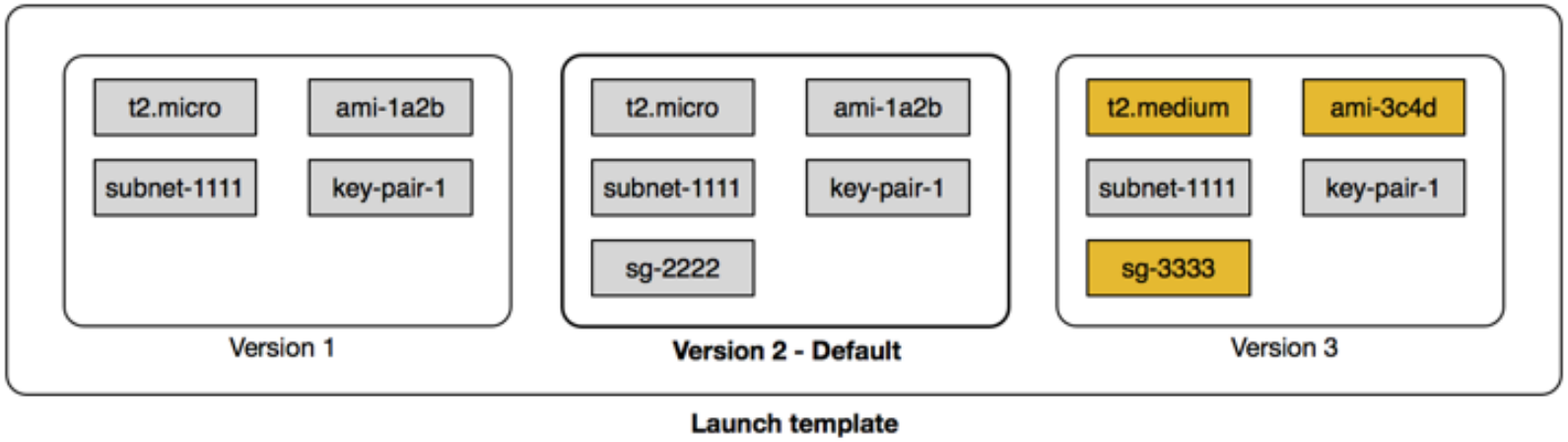




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EC2 Launch Template

Launch Template Versions



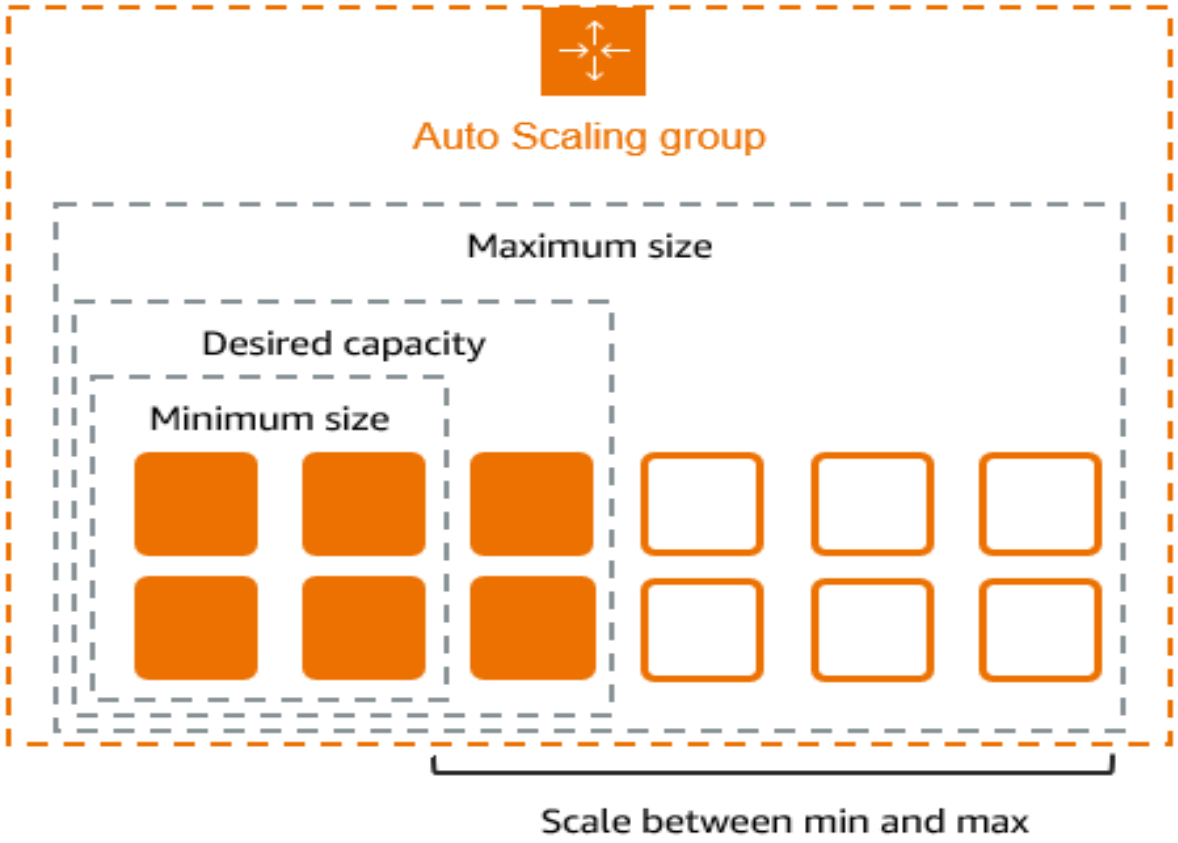
- Each version has a number and cannot be edited after creation.
- To change a setting, create a new version from an existing one.
- $Default is used when no version is selected; $Latest may differ.

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EC2 Launch Template

Amazon EC2 Auto Scaling

- Auto Scaling Groups manage the number of EC2 instances.
- A Launch Template provides instance settings.
- Select a template version when creating the group.
- Scaling launches new instances; it does not rewrite running ones.



EC2 Launch Template

Advantages of Launch Templates

- Repeatable EC2 configurations for teams and labs.
- Version changes without overwriting earlier versions.
- Supports Auto Scaling and a wide range of EC2 features.
- Reduces configuration mistakes—but permissions still matter.

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EC2 Launch Template

How to Create a Launch Template

1. EC2 Console → Launch Templates → Create launch template.
2. Name it and choose an Amazon Linux 2023 AMI.
3. Select a compatible instance type, key pair and Security Group.
4. Add optional storage, tags and user data.
5. Create version 1 → Launch an EC2 instance from it.
6. Create version 2/3 when configuration changes.

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EC2 Launch Template

Launch Templates vs. Legacy Configurations

Launch Templates (recommended)

✓ Numbered versions

✓ EC2 + Auto Scaling uses

✓ Modern instance features

✓ Use for all new labs

Launch Configurations (legacy)

• No versioning

• Used only with Auto Scaling

• Limited newer instance support

• Not available to new accounts

New accounts created on/after Oct 1, 2024 cannot create launch configurations.

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EC2 Launch Template

Three Versions in Our Hands-On Lab

Version 1 — Base

Amazon Linux 2023

EC2 + SSH

No NGINX yet

Version 2 — NGINX

Install NGINX by user data

Start the service

Show welcome page

Version 3 — Website

Install NGINX + Git

Deploy Techpro designer site

Show custom page

Each version starts a NEW instance; running EC2 instances are not automatically updated.

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EC2 Launch Template

Security and Cost Checklist

Security

SSH / TCP 22 → My IP (/32)

HTTP / TCP 80 → Anywhere (public web lab)

Use own SSH key; never share private key

Do not use chmod 777

Cost and Network

Pick public subnet and public IPv4 if needed

Check instance size and EBS costs

Watch public IPv4 charges

Terminate lab EC2 instances afterward

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EC2 Launch Template

Validate, Troubleshoot and Clean Up

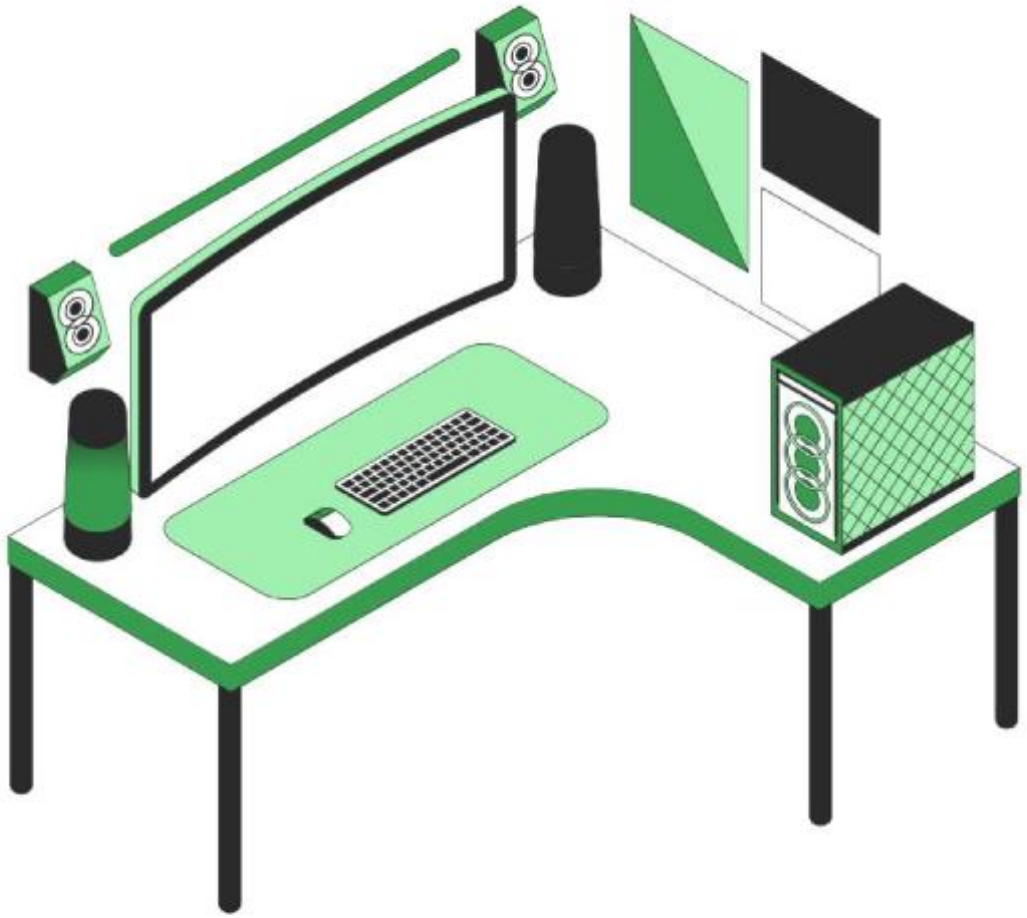
Test Each New Instance

- EC2 status checks: 2/2 passed
- SSH into the instance
- sudo systemctl status nginx
- curl -I http://localhost
- Open http://PUBLIC-IP in browser

If Something Fails / Cleanup

- Check TCP 80 in Security Group
- Read /var/log/cloud-init-output.log
- Verify correct template version
- Terminate V1/V2/V3 EC2 instances
- Delete template and leftover lab resources

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Do you have any questions?

Send it to us! We hope you learned something new.

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